Motor Coordination: A Local Hub for Coordination

A local interneuron of a crayfish central pattern generator serves as a hub that integrates ascending and descending coordinating information and passes it on to a local oscillatory microcircuit to coordinate a series of segmental appendages known as swimmerets.

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Have you every observed an eel snake through the water or a centipede scuttle, not to mention a horse gallop, and wondered how they coordinate the movements of body segments or limbs to produce the functional patterns that propel them? The problem of intersegmental coordination must be solved by the nervous system across metazoans that are segmented or limbed. In many such cases of locomotory movements, axial muscles or limbs along the body must move in a sequence such that, in each cycle of the rhythmic movement, the sequence must be proportionally maintained regardless of how fast the cycle proceeds. Technically, we say such movement patterns are phase constant: the delay between segmental movements in the sequence is a constant fraction of the cycle. A new study by Smarandache-Wellmann *et al.* [1] addresses the cellular mechanisms for such coordination.

This problem has been addressed by neuroscientists in many different animals from mice stepping with a coordinated gait, to fish undulatory swimming, to flying locusts which must coordinate their two pairs of wings. Crayfish and lobsters have caught the imagination of neuroscientists interested in this problem, because of their historical importance in laying down the foundations of the concept of the central pattern generator [2]. These animals have, on their abdominal segments A2–A5, four pairs of appendages called swimmerets that can beat rhythmically, helping the animal scuttle 'across the floors of silent seas'. The swimmerets in each segment are synchronous in their beat, but between segments there is a rear-to-front delay of 0.25 of a cycle (that is, a phase delay of 0.25).

The isolated abdominal nervous system of the crayfish, consisting of segmental ganglia A1–A6, can produce, when command neurons are activated or stimulated pharmacologically with carbachol, a coordinated pattern of activity in power stroke and return stroke motor neurons for each pair of swimmerets that mirrors what is observed in intact animals (Figure 1) [3]. The neuronal networks that produce this fictive central pattern are bilaterally repeated, allowing us to focus on one side of the neuraxis only, and Mulloney, Smarandache-Wellmann and their co-workers [3] have leveraged this historic preparation to pursue a cellular description of how phase constant intersegmental coordination is accomplished in the nervous system.

The general and detailed structure of this network is known (Figure 1) [3]. Each segment contains neuronal microcircuits which produce antiphasic oscillations in power stroke and return stroke non-spiking interneurons, generated by mutual inhibition, which program power stroke and return stroke motor activity in that segment. These oscillatory segmental microcircuits are bilaterally linked and can be conveniently termed segmental oscillators. The problem then reduces to that of coordinating these oscillators.

For several years now it has been known that coordinating information is sent from each segmental oscillator to neighboring segmental oscillators, both frontward (ascending coordinating neurons) and rearward (descending coordinating neurons) [4–6]. These projection interneurons are inhibited respectively by the power stroke and return stroke interneurons of the segmental oscillators, so that the ascending coordinating neurons fire during the return stroke and vice versa for the descending coordinating interneurons (Figure 1). Thus, in segment A4 information is collected from ascending coordinating interneurons from A5 and descending coordinating interneurons from A2 and A3 (Figure 1). These coordinating interneurons can, if their activity is perturbed, influence the phase of their target segmental oscillators. In this process, information is collected from the coordinating interneurons in each hemi-segment by a non-spiking commissural interneuron called ComInt1 [5,6]. How, then, does this commissural interneuron influence its segmental oscillator?

The answer to this question is elegantly provided in the new study by Smarandache-Wellmann *et al.* [1]. ComInt1 is strongly electrically coupled to the return stroke interneurons of the segmental oscillator, so that information from ascending and descending coordinating interneurons in the form of excitatory synaptic potentials is integrated in ComInt1 and passed on to the segmental oscillator to coordinate its phase with that of the neighboring segmental oscillators. At that point, each subsequent synapse within the segment is inhibitory, with the power stroke oscillator interneurons inhibiting descending coordinating interneurons and power stroke motor neurons, and return stroke oscillator interneurons inhibiting ascending coordinating interneurons and return stroke motor neurons (Figure 1).

Because the connections are inhibitory it is somewhat difficult to follow the activity, but the net effect is that, for example, ascending coordinating interneurons and return stroke motor neurons fire when return stroke oscillator interneurons are inhibited. A key finding is that perturbations of the ComInt1 reset oscillations in the same and neighboring segments in a manner commensurate with a 0.25 phase difference between segments. Thus, an important mechanism for intersegmental coordination of oscillatory microcircuits is established at the cellular level, and in this mechanism ComInt1 serves as an integrating node or hub.

Smarandache-Wellmann *et al.* [1] speculate that an integrating node like ComInt1 may also operate in less well understood systems [7–11], for example, those that program undulatory swimming in fishes [12]. There are, however, few systems where enough cellular detail is known to look properly for signs of the newly defined mechanism. The

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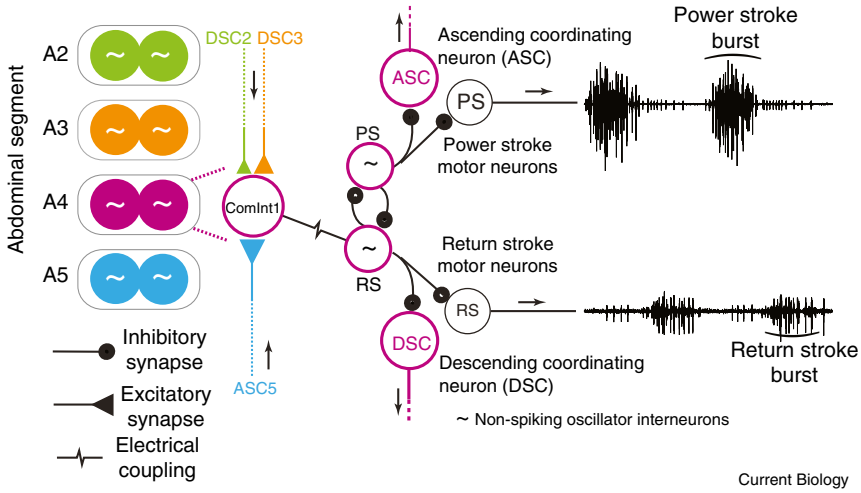


Figure 1. A cartoon of the organization of segmental oscillators that produce rhythmic swimmeret movements in the abdominal nerve cord of the crayfish (segments A2–A4).

The segmental microcircuits are bilateral and sufficiently linked within a segment that one side only can be considered. Such hemisegmental circuits are illustrated for segment A4, including the integrating hub ComInt1, the hemisegmental oscillator consisting of power and return stroke interneurons, and the ascending and descending coordinating interneurons emanating from and impinging on segment A4. Also shown are the power stroke and return stroke motor neurons is the motor output recorded in corresponding motor nerves. (Adapted with permission from the Society of Neuroscience [1].)

species for which we have the most relevant knowledge is the leech [13,14]: in the leech, quasi-independent segmental oscillator microcircuits of interneurons produce an undulatory wave that propels swimming, and the system is phase constant (albeit with a phase set point that reflects the viscosity of the medium) [12,13]. In this system, the interneurons that make up the segmental oscillators are themselves projection neurons that ascend and descend to neighboring segments. The system of oscillatory microcircuits in the leech is thus highly concatenated and it is hard to differentiate the coordinating function of each interneuron from its role in generating the oscillations themselves; there are neither independent coordinating neurons nor specific integrating nodes for coordinating information. Moreover, sensory feedback is critical in establishing a match between the undulatory wave and the fluid medium [15].

Leech circulation is propelled by peristaltic waves in longitudinal tube-like hearts that are controlled by motor neurons in 16 contiguous segments [13]. Crucial to this coordination are two oscillatory microcircuits based on mutual inhibition between bilateral oscillator interneurons that reside in the third and fourth segments. These segmental oscillators are coordinated by two bilateral pairs of independent coordinating interneurons, which are not themselves oscillatory but are rhythmically inhibited by the oscillator interneurons [16]. The oscillator interneurons and the coordinating interneurons mutually inhibit one another ipsilaterally, and the net effect is coordination between the two segmental oscillators that flexibly determines phase based on which of the two segmental oscillators has the fastest period [17–19]. Thus, in this system, as in the case of the crayfish swimmeret, there are segmental oscillators that rhythmically inhibit independent coordinating neurons, but the coordinating neurons themselves act as the integrating nodes for communication between the oscillators and oscillator phasing is flexible.

It is clear from the examples discussed that there are different mechanisms that can effect intersegmental coordination of oscillatory microcircuits and, with each example where we gain a hard-won understanding, as in the one studied by Smarandache-Wellmann *et al.* [1], we are beginning to gain insights into how mechanisms are matched with the functional demands of the behavior controlled.

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